

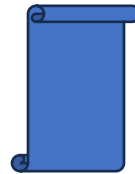
RETAIL SALES ANALYSIS DASHBOARD



◆ PROJECT OVERVIEW

This project demonstrates an end-to-end retail sales analytics workflow, beginning with data exploration and cleaning in **Python** and progressing through **data transformation, feature engineering, data modeling, DAX development and interactive dashboard creation in Microsoft Power BI**. The objective was to transform a raw and inconsistent retail sales dataset into a reliable analytical dataset that could be used to **evaluate sales performance, customer purchasing behavior, transaction patterns, payment preferences, purchasing channels and sales trends over time**. The overall workflow followed a structured analytics process:

Raw Dataset → Data Exploration → Data Cleaning → Data Transformation → Feature Engineering → Data Export → Power BI Data Modeling → DAX Measures → Interactive Dashboard → Business Insights



◆ THE PROBLEM STATEMENT:

The Stakeholders asked to develop an interactive retail analytics solution that enables Sales team to understand how **customers transact, what they purchase, how much they spend, which categories contribute to revenue and how sales performance changes over time**. The dashboard was designed to answer key business questions such as **how much revenue was generated, how many transactions occurred, how many unique customers were served, what the average quantity purchased per transaction was, what the average price per transaction was, which customers demonstrated the strongest purchasing activity, which categories generated the highest transaction volume and revenue, which payment methods were most frequently used, how online and in-store transactions compared, which weekdays experienced the highest transaction activity and how current sales performance compared with the previous year**. The overall goal was to create a centralized and interactive reporting solution that allows users to move **from high-level KPIs into detailed customer, category, payment, channel and time-based analysis**.



◆ DATASET DETAILS

The dataset used for this project was sourced from **Kaggle** and is titled **Retail Store Sales - Dirty Dataset**. The dataset was particularly suitable for demonstrating a practical data analytics workflow because it contained several **real-world data quality issues that required investigation and correction** before the information could be reliably analyzed. The original dataset contained approximately **12,576 transaction records** and included information related to **transactions, customers, products, prices, quantities, total spending, payment methods, transaction locations, transaction dates and discount information**. The raw data contained **missing values, duplicate records, inconsistent category values, incorrect or incomplete transaction totals, data type mismatches and invalid or inconsistent entries**, providing an opportunity to demonstrate how data quality can directly influence business reporting and analytical outcomes.

The dataset consisted primarily of transaction-level retail information. Each transaction was identified using a unique Transaction ID, while Customer ID was used to identify individual customers, with **approximately 25 unique customers** represented in the dataset. The Category field represented the purchased **product category, while the Item field contained the name or identifier of the purchased product and could contain missing values**. The dataset also included **Payment Method, which captured how customers paid for their purchases**, which represented the **transaction channel such as online or in-store**. Transaction Date recorded when the purchase occurred, while Discount Applied indicated whether a discount was associated with the transaction, although this field contained a substantial number of missing values.



◆ DATA EXPLORATION AND CLEANING

The data preparation stage was performed using Python, specifically Pandas within a Jupyter Notebook environment. **The first step involved exploring and profiling the raw dataset to understand its structure, size, data types, statistical characteristics, and potential quality issues.** Functions such as **.info()**, **.describe()**, **.shape**, **.isnull()** and **.duplicated()** were used to examine the dataset and identify missing values, duplicates, inconsistent data types and potential anomalies. Duplicate records were identified and removed where appropriate to ensure that the cleaned dataset provided a more reliable representation of the underlying

transactions. This initial exploration provided a clear understanding of the data quality challenges and helped establish an appropriate cleaning strategy before the dataset was introduced into Power BI. This all steps was important because the dashboard calculations were dependent on accurate transaction-level data.

- ❖ **Missing Value Treatment** : The **Discount Applied** column contained a large number of missing values and because there was insufficient reliable information to determine whether **a discount had actually been applied, the column was removed instead of introducing potentially inaccurate assumptions through imputation.** For numerical fields such as Quantity, Price Per Unit and Total Spent, logical relationships between the fields were used to recover missing information where possible.
- ❖ **When Quantity and Price Per Unit were available but Total Spent was missing,** the transaction amount was recalculated using the relationship **Total Spent = Price Per Unit × Quantity.** Similarly, **when Total Spent was available but either Quantity or Price Per Unit was missing,** the missing value was derived mathematically where sufficient information was available. Quantity could be calculated **using Total Spent ÷ Price Per Unit, while Price Per Unit could be calculated using Total Spent ÷ Quantity.** This approach helped preserve useful transaction records while maintaining consistency between related numerical fields.
- ❖ **Data Standardization and Transformation** : The dataset also required standardization of categorical information. Inconsistent category values were identified and corrected so that **logically identical categories would be represented consistently.** This was important because inconsistent category labels could cause the same product category to appear as multiple groups in Power BI, **resulting in inaccurate category-level transaction and revenue analysis.** Data types were also converted into appropriate formats.
- ❖ **Column names were standardized to improve readability and maintain consistency** throughout the analytical workflow. Clear and standardized field names made the dataset easier to understand and simplified the development of DAX measures and Power BI visuals.



◆ WHAT'S INSIDE

The final Power BI dashboard brings together the analytical components into a single interactive reporting experience.

 **Revenue Performance** : Total Revenue, Previous-Year Revenue, Year-over-Year Sales Growth and Transaction volume

 **Customer Insights** : Best Customer identification, Customer transaction activity, Average purchase quantity, Average purchase value and Customer-level revenue contribution

 **Category Analysis:** Transactions by category, Revenue by category and Category-level purchasing behavior

 **Payment & Channel Analysis** : Payment-method distribution, Online vs. in-store performance and Transaction behavior by purchasing channel

 **Time-Based Analysis** : Yearly sales trends, Transaction trends, Weekday transaction patterns and Year → Quarter → Month → Day hierarchy. Together, these views provide both a **high-level management perspective** and the ability to explore specific dimensions of business performance.



◆ DAX FORMULAS & NEW MEASURES

A major component of this Power BI project was the development of **DAX measures and calculated date logic** to transform raw transactional data into meaningful business KPIs. Rather than relying only on basic aggregations, I created measures that support customer analysis, revenue performance, transaction behavior, and year-over-year comparisons.

Key Measures & KPIs Created

- **Total Revenue** – Calculates the overall revenue generated across all transactions.
- **Sales Previous Year** – Calculates prior-year sales to establish a reliable comparison point.
- **Sales Growth %** – Measures year-over-year sales growth and helps identify whether revenue performance is improving or declining.
- **Total Transactions** – Provides the overall transaction volume and supports analysis of purchasing activity.
- **Total Customers** – Calculates the unique customer base, with the dashboard showing **25 customers**.
- **Average Price per Product/Transaction** – Helps understand the typical monetary value associated with purchases.
- **Average Quantity per Transaction** – Measures the average number of products purchased in each transaction.
- **Best Customer Analysis** – Identifies the highest-value customer based on transaction activity and purchasing behavior.
- **Date Hierarchy & Time Intelligence** - To support time-based analysis, I created a structured date hierarchy containing **Year → Quarter → Month → Day**

- I also developed date-table calculations to return the **day number and weekday number**, allowing transactions to be analyzed according to specific days of the week. This structure made it possible to move seamlessly from high-level annual performance to more detailed monthly, daily, and weekday-level analysis.



◆ VISUALS UTILIZED TO ANALYZE THE DATA

- A dedicated Best Customer KPI was created to identify the highest-performing customer based on the customer analysis logic implemented in the dashboard include approximately 13K total transactions, an average quantity per transaction of 6, and an average price per transaction of 23.37. Rather than displaying only the customer's name, the KPI provides supporting performance indicators that allow users to understand the customer's transaction activity and purchasing characteristics.

Revenue & Transaction Analysis

A **Line + Column Chart** was used to analyze **Total Revenue and Transactions by Customer**. This visualization makes it easier to compare customer contribution while simultaneously understanding transaction volume. It helps identify whether high revenue is driven by frequent purchases, higher-value transactions, or a combination of both. The analysis also supports the customer-ranking logic used to identify the Best Customer displayed in the KPI card.

Transactions by Weekday

A **Tree map** was implemented to analyze transaction distribution across weekdays. This provides a quick visual understanding of which days generate higher transaction activity and helps identify recurring purchasing patterns.

Category Performance

Two **Bar Charts** were incorporated to analyze Transactions by Category and Revenue by Category. Using both measures provides a balanced view of category performance. A category may generate a high number of transactions without necessarily generating the highest revenue, making it important to evaluate **volume and financial contribution together**.

Payment Distribution

A **Donut Chart** was used to illustrate payment-method distribution, including methods such as Cash, Credit Card and Other available payment methods. This allows stakeholders to quickly understand customer payment preferences and the relative contribution of each payment method.



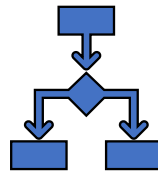
Online vs. In-Store Transactions

A second **Donut Chart** was developed to compare transaction modes Online and In-store. This provides a straightforward view of purchasing channels and helps highlight how customers interact with the business.



Sales & Transactions Over Time

A **Line Chart** was created to examine **Sales and Transactions over time**, with the analysis structured around yearly trends. This visualization supports year-over-year performance evaluation and complements the DAX-based **Sales Growth %** and **Previous Year Sales** measures.



◆ STEPS I FOLLOWED TO BUILD THE DASHBOARD



Understanding the Data & Business Requirements: Based on the data, I identified the key areas that needed to be explored, including **revenue, transactions, customers, product categories, payment methods, purchasing modes and time-based trends**. This helped me define the KPIs and visuals required to create a dashboard that would provide both a high-level performance overview and deeper business insights.



Building Customer Analysis: I then focused on understanding **customer purchasing behavior and identifying the best-performing customer**. Rather than evaluating customers using only revenue, I used multiple measures to provide a broader perspective of **transaction frequency, purchasing quantity, and average transaction value**.



Creating KPI Cards: After developing the DAX measures, I created **KPI cards to provide an immediate snapshot of business performance**. The KPI section highlights important metrics such as **Total Revenue, Sales Growth %, Previous-Year Sales, Total Transactions, Total Customers, and Best Customer performance**. These KPIs were positioned prominently so users could quickly understand the overall business performance before exploring the detailed visualizations.



Organizing the Dashboard & Adding Analytical Flow : Once the individual visuals were created, I organized them into a **logical and user-friendly dashboard layout**. The analytical flow was designed to move from **overall KPIs → customer performance → category analysis → payment and transaction modes → time-based trends**. This structure allows users to begin with the overall business picture and gradually explore the factors influencing revenue and transaction performance.



◆ UI DESIGN AND THEME COLORS

The dashboard design was developed with a strong focus on visual consistency, readability, and user experience. Since the purpose of a business dashboard is not only to display information but also to communicate it efficiently, the visual design was kept clean and structured.

● Green Theme

The primary dashboard view uses a green-based color theme to create a consistent visual identity throughout the report. Green was used across key dashboard elements to establish visual continuity while helping important KPIs and analytical sections stand out without creating excessive visual noise.

🌸 Pink Alternative View

A pink color theme was also incorporated as an alternative visual treatment for the same dashboard.



◆ KEY INSIGHTS

📊 Overall Analysis Context

The key insights were derived with the **Payment Method and Category filters set to “All”**, while the analysis period was defined from **January 1, 2022, through January 8, 2025**. This provides an overall view of **customer activity, revenue performance, transaction behavior, payment preferences, purchasing channels, and time-based trends** without restricting the analysis to a specific category or payment method. The dashboard therefore provides a **consolidated perspective of the retail business** and highlights important patterns across customers, products, transaction channels, payment methods, and time.

💰 Revenue Performance

During the selected analysis period, the business generated approximately **\$1.69 million in total revenue**, demonstrating the overall financial contribution of the transactions captured in the dataset. Compared with **previous-year sales of approximately \$1.15 million**, the business recorded a **47.90% Sales Growth**. This represents a strong year-over-year improvement in revenue performance and indicates meaningful growth during the analyzed period.

Best Customer – Emma

The customer analysis identifies **Emma as the most valuable customer**, based on the highest revenue and transaction contribution within the analyzed dataset. Emma is associated with approximately **13K total transactions**, an **average quantity of 6 units per transaction**, and an **average price per product of \$23.37**. These metrics indicate a high level of purchasing activity and position Emma as a **significant contributor to overall retail performance**. Identifying high-value customers can help businesses better understand customer behavior.

Category Performance

Food emerged as the leading product category, ranking highest in both **transaction volume and revenue contribution**. This indicates that Food products represent a major component of customer purchasing activity as well as overall sales performance.

The category's strong position across both measures is particularly important because **transaction volume and revenue do not always move together**. In this analysis, Food demonstrates strong performance from both a **purchasing-frequency and financial-contribution perspective**.

Online vs. In-Store Transactions

The channel analysis shows that purchasing activity is **relatively balanced between online and in-store channels**, with online transactions having a modest lead.

- **Online Transactions: 6,354**
- **In-Store Transactions: 6,221**

The relatively small difference indicates that customers are actively using **both purchasing channels** rather than relying exclusively on online or physical-store transactions. This balanced channel performance provides useful context for **channel management, customer experience, and sales strategy**.

Payment Method Distribution

The payment-method analysis shows that **Cash is the most frequently recorded payment method**, followed closely by Digital Wallet and Credit Card.

-  **Cash: 4,310 Transaction IDs**
-  **Digital Wallet: 4,144 Transaction IDs**
-  **Credit Card: 4,121 Transaction IDs**

Although Cash has the highest transaction count, the relatively narrow difference between the three methods indicates that **customer payment preferences are distributed across multiple payment options**. Digital Wallets and Credit Cards also represent substantial transaction activity, demonstrating the importance of **electronic payment methods alongside traditional cash payments**.

Weekday Transaction Activity

The weekday analysis indicates that **transaction volume is generally higher during weekdays than weekends**. This suggests that customer purchasing activity is more concentrated during the working week within the analyzed dataset.

This pattern can provide useful context for **staffing, inventory planning, promotional timing, and customer engagement initiatives**. Higher weekday activity may also indicate that the purchasing behavior of the customer base is more closely associated with weekday activity.

Customer & Transaction Activity

The dashboard represents **25 unique customers** and approximately **12K+ transactions** across the selected period. The combination of customer count and transaction volume provides an indication of how frequently customers interact with the business.

The **Best Customer analysis further demonstrates that transaction activity is not evenly distributed across all customers**, with Emma standing out as a particularly active and valuable customer.

Year-over-Year Growth

The comparison between current and previous-year revenue demonstrates a **positive year-over-year performance trend**. Total revenue of approximately **\$1.69 million**, compared with previous-year sales of approximately **\$1.15 million**, resulted in a reported **47.90% Sales Growth**. The dashboard also enables **transaction trends to be evaluated alongside revenue trends**, providing a broader understanding of whether changes in financial performance are accompanied by changes in transaction activity. This is important because revenue growth can be influenced by both **transaction volume and the value of individual purchases**.

Overall Business Interpretation

The business generated approximately **\$1.69 million in revenue**, representing **47.90% growth compared with the previous year**. **Food remained the strongest category** in terms of both transactions and revenue, while **Emma was identified as the most valuable customer** based on the implemented customer-performance analysis. Transaction activity was relatively balanced across purchasing channels, with **6,354 online compared with 6,221 in-store transactions**. In terms of payment preferences, **Cash led with 4,310 transactions**, followed by **Digital Wallet with 4,144** and **Credit Card with 4,121**. The analysis also showed that **weekday transaction activity exceeded weekend activity**. These findings demonstrate that the business experienced **strong revenue growth while maintaining substantial activity across multiple customer channels and payment methods**.



◆ THE STORYTELLING

 **Business Analysis Narrative** : The analysis begins with the overall financial picture. During the selected period of **January 1, 2022, to January 8, 2025**, with **Category and Payment Method filters set to All**, the business generated approximately **\$1.69 million in total revenue**. Compared with **\$1.15 million in previous-year sales**, this resulted in a reported **47.90% sales growth**, establishing the central insight that the retail business experienced **strong revenue growth**.

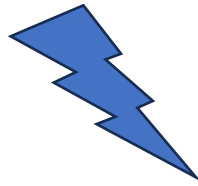
Among the **25 customers** represented in the dataset, **Emma emerged as the highest-performing customer** based on the implemented customer analysis. She contributed approximately **13K transactions**, with an **average quantity of 6 units per transaction** and an **average price per product of \$23.37**. This demonstrates how customer-level analysis can move beyond overall sales to identify **highly active and valuable customers**.

The analysis then moves to product performance, where **Food emerged as the leading category** in both **transaction activity and revenue contribution**. This indicates that Food is not only frequently purchased but also makes a significant financial contribution to the business, providing a more complete view of category performance.

The dashboard also examines **where and how customers make purchases**. Online and in-store activity remained relatively balanced, with **6,354 online transactions** compared with **6,221 in-store transactions**. Payment behavior was similarly distributed, with **Cash leading at 4,310 transactions**, followed by **Digital Wallet at 4,144** and **Credit Card at 4,121**. These results show that customers actively use **multiple purchasing channels and payment methods**.

Finally, the analysis examines **when customers transact**. **Weekday transaction activity was generally stronger than weekend activity**, while the time-series analysis highlighted **positive sales and transaction performance over time**. Overall, the dashboard tells a clear business story: **strong financial growth supported by active customers, a high-performing Food category, balanced purchasing channels, diverse payment preferences, and stronger weekday activity**.

The dashboard therefore moves users from the broad question of **“How is the business performing?”** to deeper questions such as **“Who is driving performance?”**, **“What are customers purchasing?”**, **“How are they paying?”**, **“Where are they purchasing?”**, and **“How is performance changing over time?”**.



◆ CHALLENGES & HOW THEY WERE ADDRESSED

The original dataset contained **missing values, inconsistent category names, duplicate records, incorrect or incomplete transaction totals and data type mismatches**. These issues could directly affect important Power BI metrics such as **Total Revenue, Total Transactions, Average Quantity, and Average Price**. To identify these issues before building the dashboard, I performed data profiling in **Python using Pandas**, including `.info()`, `.describe()`, `.shape()`, missing-value checks, and duplicate analysis.

✍ Handling Missing & Inconsistent Values

Handling missing numerical values required careful consideration because removing every incomplete record could result in **unnecessary data loss**. Where possible, I used the mathematical relationship between **Price Per Unit, Quantity, and Total Spent** to derive missing values. Remaining missing Quantity values were addressed using the **mode**, helping preserve usable transaction records while maintaining numerical consistency. The **Discount Applied** field contained a substantial number of missing values with no reliable basis for determining whether a discount had actually been applied. Rather than introducing unsupported assumptions, I **removed the field from the final analytical dataset**, following the principle of avoiding unreliable imputation.

📄 Standardizing Categories & Data Types

The dataset contained **inconsistent category values**, which could cause identical categories to appear as separate groups in Power BI. I standardized these values during the Python cleaning process to ensure reliable **category-level filtering, revenue analysis, and transaction comparisons**. I also corrected **data type mismatches**, converting Transaction Date into an appropriate date format and ensuring **price, quantity, and transaction values were stored as numerical fields**.

📅 Creating Time-Based Analysis

Another challenge was developing meaningful comparisons over time. I created a date structure supporting **Year → Quarter → Month → Day**, along with **Month Name, Day Name, and Weekday Number** for chronological sorting and weekday analysis. DAX measures such as **Previous-Year Sales and Sales Growth %** were then developed to compare current performance with historical results.

Managing Dashboard Complexity

With multiple analytical dimensions including **revenue, growth, customers, categories, payments, transaction channels, weekdays and time trends** another challenge was presenting the information without making the dashboard visually overwhelming. To address this, I selected visualizations based on their analytical purpose: **KPI cards for headline metrics, bar charts for category comparisons, donut charts for payment and channel distribution, a tree map for weekday activity, and line charts for time-based trends.** Slicers were also incorporated to give users control over the analysis while keeping the dashboard clean and interactive.



◆ WHY THIS PROJECT MATTERS

✚ This project demonstrates that **effective business decisions depend on reliable and well-prepared data.** A visually appealing dashboard alone cannot provide meaningful insights if the underlying data contains **missing values, duplicates, inconsistent categories, incorrect totals, or inappropriate data types.** Therefore, the project follows the beginning with raw data quality challenges and using **Python and Pandas to explore, clean, transform and validate the dataset** before using it for Power BI reporting.

🔗 The project also highlights the connection between **data preparation and business intelligence.** Python was used to address data-quality and transformation requirements at the source level, while **Power BI converted the prepared data into KPIs, DAX measures, time-based analysis, and interactive visualizations.**

 From a business perspective, the dashboard brings multiple dimensions of retail performance into **one interactive reporting environment.** Instead of analyzing the fields separately, stakeholders can evaluate them together. For example, the dashboard highlights approximately **\$1.69 million in revenue and 47.90% sales growth**, while also providing context around the customers, categories, channels and payment methods contributing to that performance.

🎯 The project further demonstrates how data can support **targeted business decisions.** Identifying **Emma as the highest-performing customer** can support customer retention and engagement strategies, while recognizing **Food as the leading category** can inform inventory planning, promotions and sales strategy. Similarly, analyzing **online vs. in-store transactions** and **payment-method preferences** provides useful insights for channel management and customer experience.



STRATEGIC RECOMMENDATIONS

 **Sustain & Understand Revenue Growth** : The business should continue monitoring the drivers behind the strong **47.90% sales growth** and approximately **\$1.69 million in total revenue**. Further analysis should determine whether growth is primarily driven by **transaction volume, average transaction value, category performance, or customer purchasing behavior** so successful strategies can be maintained and expanded.

 **Strengthen the Leading Food Category** : Since **Food is the leading category in both transactions and revenue**, the business should prioritize **product availability and inventory planning** for this category. Further product-level analysis can identify top-performing items and explore **cross-selling, bundling, and complementary-product opportunities** to increase transaction value and improve lower-performing categories.

 **Focus on Customer Retention** : The identification of **Emma as the most valuable customer** highlights the importance of customer segmentation and retention. Customers can be segmented using **transaction frequency, total spending, average quantity, and average purchase value**. High-value customers could then be targeted through **loyalty programs, personalized promotions, exclusive offers, and retention initiatives**.

 **Optimize Online & In-Store Channels** : With **6,354 online transactions** and **6,221 in-store transactions**, purchasing activity is relatively balanced across both channels. The business should continue supporting **both online and in-store experiences** while comparing revenue, average transaction value, product preferences, and customer behavior across channels to develop more targeted strategies.

 **Maintain Flexible Payment Options** : Payment activity is distributed relatively evenly across the major methods, with **Cash at 4,310 transactions, Digital Wallet at 4,144, and Credit Card at 4,121**. The business should continue offering **multiple payment options** while monitoring the growth of digital payments and evaluating opportunities for payment-specific promotions or loyalty benefits.

 **Improve Weekend Performance** : Since **weekday transaction activity is higher than weekend activity**, the business could explore **weekend promotions, special offers, product bundles, and digital campaigns** to increase activity during lower-volume periods. Further analysis should identify the specific days with the highest and lowest transaction volumes before implementing targeted initiatives.

 **Continue Monitoring Year-over-Year Trends** : The business should continue using **year-over-year and time-series analysis** to monitor whether the positive growth trend remains consistent, accelerates, or begins to slow. Breaking these trends down by **category, customer, channel, and payment method** can help identify emerging changes in business performance.

 **Expand the Dashboard for Advanced Analytics** : Future versions of the Power BI dashboard could incorporate **customer segmentation, product-level analysis, profitability measures, average basket value, repeat-customer analysis, geographic analysis, and forecasting**. These enhancements could evolve the dashboard from a descriptive reporting solution into a more comprehensive **decision-support and predictive analytics tool**.

 **Overall Strategic Direction** : Overall, the business should focus on **sustaining revenue growth, strengthening customer retention, supporting the high-performing Food category, optimizing online and in-store channels, monitoring payment preferences, and improving lower-volume weekend activity**. Continued use of data-driven analysis will help ensure that decisions consider not only overall revenue but also the **customer, product, transaction and channel behaviors driving business performance**.